POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2021
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

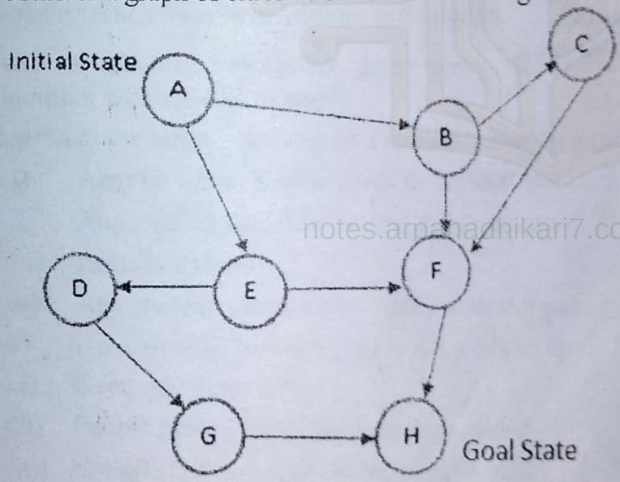
a) What is Artificial Intelligence? Discuss the impact of artificial intelligence on society. 7

b) What is an intelligent agent? Describe the task environment of an autonomous car. 8

a) What are the components of a state space search problem formulation? Formulate the 8-queens problem. 7

b) How does the interactive deepening search resolve the problem of depth-first search and breath- first search? Explain with an example. 8

a) Consider a graph of cities as shown in below figure: 7



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i. Represent state space of this graph using prolog.

ii. Write a simple search algorithm in prolog to reach to the goal state from initial state.

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b) Represent the following facts in predicate logic:
   i. John likes all kind of food.
   ii. Apple and vegetable are food.
   iii. Anything anyone eats and not killed is food.
   iv. Anil eats peanuts and still alive.
   v. Harry eats everything that Anil eats.

4. a) Consider the facts given in Q.3 b) using resolution prove that John likes peanuts.
   b) What is machine learning? Discuss the difference between regression, classification and clustering in machine learning.

5. a) Discuss Naive Bayes' algorithm with an example.
   b) How can we find the right hyperplane in SVM? Using K-means clustering, cluster the following data into two clusters and show each step.
   {2, 4, 10, 12, 3, 20, 30, 11, 25}.

6. a) Draw the architecture of Expert System. How forward chaining and backward chaining assist the inference engine of Expert System?
   b) What are the different types of ambiguities that accur in Natural Language Processing? Explain with examples.

7. Write short notes on: **(Any two)**
   a) Fuzzy learning
   b) Best first search
   c) Hopfield Network

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